Major physics theories

Louis Haurie

May 15, 2026

Contents

1 Classical mechanics

1.1 Analytical mechanics

Analytical mechanics is a total reformulation of Newton's law under the work of many physicists and mathematicians like Lagrange, Maupertuis, Euler, Hamilton and Jacobi. The term "analytical mechanics" was first introduced by Lagrange as the name of his manuscript (mécanique analytique) in 1788. This work provided a general basis for all modern theories: relativity relies on Lagrangian formulation of mechanics, statistical physics uses Hamilton equations while quantum physics was initially based on Hamilton-Jacobi derivations.

1.1.1 Constraints and generalized coordinates

In Newtonian physics, a system of N points is naturally described in the ambient space. For instance in 3d, the ambient space (ie the manifold used to describe the physical space) is $\mathbb{R}^{3N}$. A state is a position vector and its associated velocity at a given time: $s = (\overrightarrow{r}, \dot{\overrightarrow{r}})$. Therefore, in 3d the corresponding space is $\mathbb{R}^{6N}$. While this certainly is correct, this isn't exactly the most adapted basis to the constraints and the geometry of the system for solving the equation of motion. For instance, a student working on the motion of a 2d pendulum often had to project forces in some direction to express the motion as a function of an angle. Even worse, when studying the friction of a box pushed horizontally, we describe the system with forces we don't even know and try to express one as a function of another. These exercises already show some weaknesses of the Newtonian approach:

- Sometimes, working in a coordinate system spanning $\mathbb{R}^3$ is unnecessary and overcomplicates the problem with redundant equations.
- Sometimes, the considered forces may be unknown, or very difficult to compute. This is particularly true for constrained force (reaction, friction...).

For this reason, the first conceptual step of analytical mechanics is the concept of generalized coordinates. Rather than trying to derive equation in the full ambient space and try to express unknown forces, we restrict ourselves only to the relevant coordinates.

More formally, in Newtonian dynamics, any system of N point like object is naturally described in ambient space ($\mathbb{R}^{3N}$ for 3d, $\mathbb{R}^{2N}$ in 2d, $\mathbb{R}^{N}$ in 1). A state is then the N positions and velocities vectors at a given time:

$$s = (\overrightarrow{r_1}, \overrightarrow{r_2}, ..., \overrightarrow{r_N}, \dot{\overrightarrow{r_1}}, \dot{\overrightarrow{r_2}}, ..., \dot{\overrightarrow{r_N}}) \tag{1}$$

Therefore, state space in Newtonian dynamics has a dimension two times bigger than the ambient space. However, this description is often redundant: many physical systems are subject to constraints that restrict the accessible configurations. For instance, a pendulum of fixed length satisfies: $x^2+y^2 = R^2$, with R the pendulum length. Thus, its motion does not explore the full plane $\mathbb{R}^2$, but only a one-dimensional curve. The coordinates of the system are linked, and the equation is called a **constraint equation**. We call a constraint equation **holonomic** if it depends only on the positions and time:

$$f(\overrightarrow{r}_1, \dots, \overrightarrow{r}_N, t) = 0 \tag{2}$$

A constraint involving velocities in a way that cannot be integrated into a position-only relation is called **non-holonomic**. The set of all holonomic constraint equations defines a subset $\mathcal{C} \subset \mathbb{R}^{3N}$ called the **configuration space** of the system. Under some regularity conditions (constraints are independent and compatible),$\mathcal{C}$ is a smooth submanifold of $\mathbb{R}^{3N}$.

Holonomic constraints are particularly convenient because they directly shrink the space of accessible configurations: each one eliminates one position variable, which is precisely what allows us

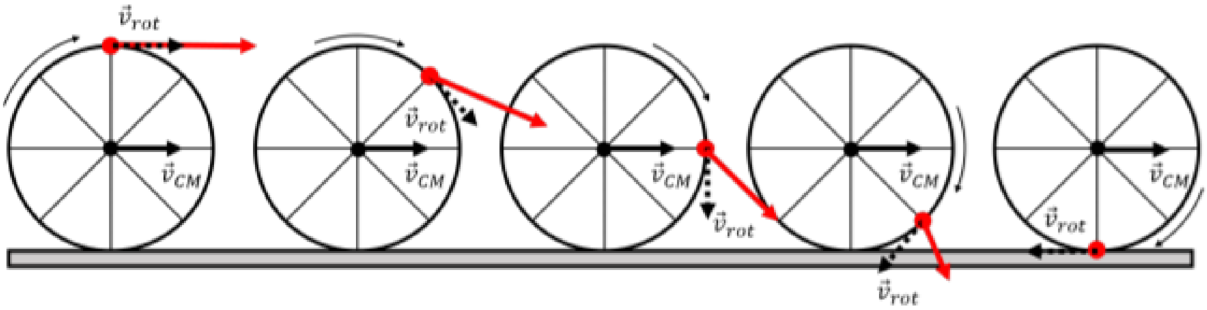

Figure 1: Example of a non-holonomic constraint: when the disk rolls without slipping on the plane, we impose the relative velocity between the disk and the plane to be zero at the contact point. Taken from phys.Libretexts.org

to pass from $\mathbb{R}^{3N}$ to the smaller manifold $\mathcal{C}$, reducing the dimension of state space and to introduce generalized coordinates as free, independent parameters.

Non-holonomic constraints do not offer this simplification. Because they constrain velocities rather than positions, they do not forbid any region of configuration space (the system can still reach any position but they restrict which directions of motion are available at each point). There is therefore no variable to eliminate, no smaller manifold to work on, and the generalized coordinate construction above simply does not apply. Concretely, this means that deriving the equations of motion may require additional tools.

A good example is a disk rolling without slipping on a plane (Fig: 1). The no-slip condition constrains the velocity of the contact point, yet the disk can reach any position (x, y) and any orientation θ on the plane: the configuration space is the full $\mathbb{R}^2 \times S^1$, unreduced. The constraint is felt not in *where* the disk can go, but in *how* it must get there.

In what follows, we restrict ourselves to holonomic constraints unless stated otherwise. Each independent holonomic constraint reduces the dimension of the accessible space by one. If the system has k independent holonomic constraints, the configuration space has dimension:

$$d = \dim(\mathcal{C}) = 3N - k \tag{3}$$

This integer d is called the number of **degrees of freedom** of the system. When $k = 0$, no constraint is imposed and $\mathcal{C} = \mathbb{R}^{3N}$: the system moves freely in the full ambient space. When $k = 3N$, the system is completely blocked at a point.

Example. A planar pendulum ($N = 1$, 2d ambient space) has one constraint $x^2 + y^2 = R^2$, so $k = 1$ and $d = 2 - 1 = 1$: a single angle suffices to describe the motion, consistent with the intuition from the introduction.

We can now give a precise definition of generalized coordinates. Since $\mathcal{C}$ is a d-dimensional smooth manifold, it can be locally parametrized by d independent real parameters $q_1, \ldots, q_d$, called **generalized coordinates**. Concretely, this means the Cartesian positions of all N particles are smooth functions of these parameters (and possibly of time, for moving constraints):

$$\vec{r}_j = \vec{r}_j(q_1, \ldots, q_d, t), \qquad j = 1, \ldots, N \tag{4}$$

By construction, any configuration satisfying all constraints can be reached by some choice of $(q_1, \ldots, q_d)$, and conversely. The q_i are free and independent: no residual constraint equation links them. The associated velocities $\dot{q}_i$ are called generalized velocities, and the $2d$-dimensional space of $(q_i, \dot{q}_i)$ is the reduced state space of the system: far smaller, in general, than the original $\mathbb{R}^{6N}$.

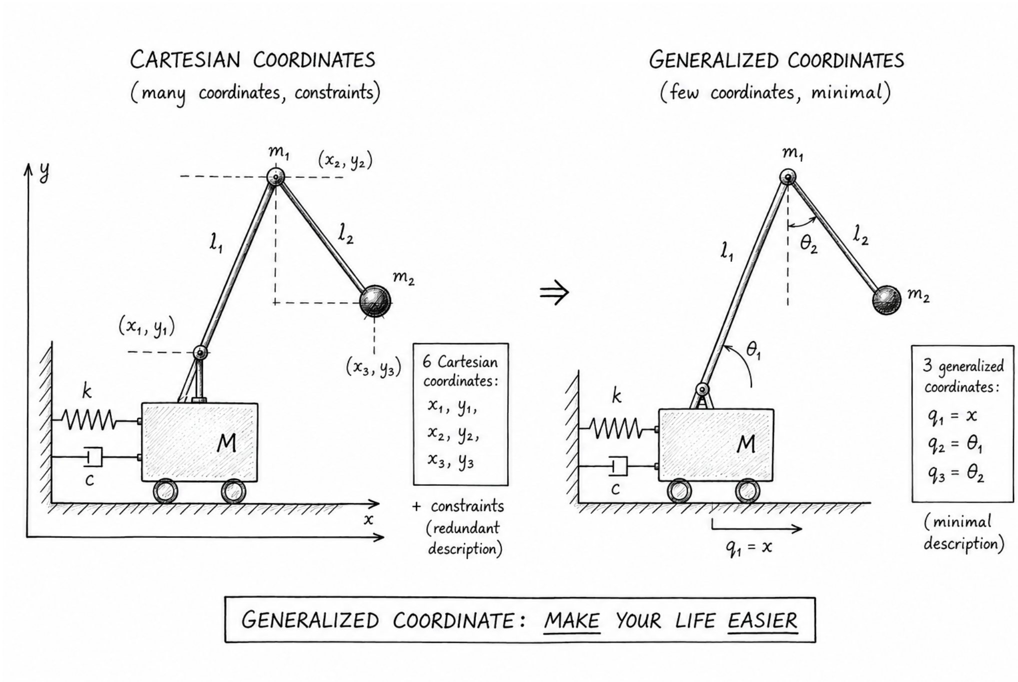

Figure 2: On complicated system such as this, the cartesian coordinates are fine but will involve a lot of projections, redundancies and reduction of the equation to arrive at a simple form. Meanwhile, working with the generalized coordinates reduces the state space dimension and ensure we are working in the right basis: minimal amount of projections and no redundancies. Illustration generated by AI - Feel free to send me yours to replace it.

In summary, generalized coordinates are not merely a clever trick: they are the natural coordinates of the configuration manifold $\mathcal{C}$, adapted to the geometry imposed by the constraints to minimize the dimension of state space. This is what makes the Lagrangian formalism both systematic and geometrically meaningful.

1.1.2 Euler-Lagrange equations

Virtual displacements

Consider a system evolving under holonomic constraints. At a given instant t, we define a **virtual displacement** $\delta\overrightarrow{r_i}$ as an infinitesimal change of the position of particle i, taken at *fixed time*, that is compatible with the constraints. In other words, if the system is at a point of $\mathcal{C}$, the displaced configuration $\overrightarrow{r_i} + \delta\overrightarrow{r_i}$ must also lie in $\mathcal{C}$.

This must be contrasted with a **natural displacement** $d\overrightarrow{r_i} = \overrightarrow{r_i}(t+dt) - \overrightarrow{r_i}(t)$, which connects two positions on the actual trajectory at two different times, and is governed by the equations of motion. Virtual displacements, by contrast, are purely geometric: they probe the constraint manifold $\mathcal{C}$ at a frozen instant, with no reference to dynamics.

Since δ and d are independent linear operations acting on separate variables (position and time respectively), they commute:

$$\delta(d\overrightarrow{r_i}) = d(\delta\overrightarrow{r_i}) \tag{5}$$

This identity has a deeper significance in field theory and quantum mechanics, where it underlies the equivalence of the Lagrangian and Hamiltonian variational principles (see the upcoming notes on quantum mechanics).

In summary, we will remember that a virtual displacement is varying the configuration (system states allowed by the constrains) while freezing the time: it is not a natural displacement, as time did not evolved under this displacement. You can picture it as a remanent image, a "imagined" state.

Virtual work and D'Alembert's principle

The constraint equations introduced earlier are purely mathematical conditions on the positions. Physically, however, something must enforce them: a hand on a table stays on the table because the table exerts some sort of force on it; a pendulum mass stays at fixed distance from the pivot because the rod pulls it back whenever it tries to stray under gravity. We call these **constraint forces** $\overrightarrow{F}_i^{\,\mathrm{c}}$ the forces that constraint the system on the configuration manifold $\mathcal{C}$.

All other forces (gravity, springs and Hooke's law, any interaction prescribed independently of the geometry...) are called **applied forces** $\overrightarrow{F}_i^{\,\mathrm{app}}$.

The total force on particle i is therefore:

$$\overrightarrow{F}_i = \overrightarrow{F}_i^{\,\mathrm{app}} + \overrightarrow{F}_i^{\,\mathrm{c}} \tag{6}$$

Constraint forces are typically unknown: we rarely know the tension in a rod or the normal force from a surface before solving the problem. This is precisely the difficulty encountered in Newtonian mechanics. However, their role gives us a crucial geometric property. Since $\overrightarrow{F}_i^{\,\mathrm{c}}$ must keep the system *on* $\mathcal{C}$ without driving it *along* $\mathcal{C}$ (that is the role of an applied force), it can only point in directions orthogonal to $\mathcal{C}$. In other words, constraint forces are normal to the constraint manifold. Since virtual displacements are by definition tangent to $\mathcal{C}$, constraint forces do no virtual work:

$$\sum_i \overrightarrow{F}_i^{\,\mathrm{c}} \cdot \delta\overrightarrow{r_i} = 0 \tag{7}$$

In other words, the virtual work is given only by the applied forces:

$$\delta W = \sum_i \overrightarrow{F}_i^{\,\mathrm{app}} \cdot \delta\overrightarrow{r_i} \tag{8}$$

This is not an approximation: it is a direct consequence of the geometry of $\mathcal{C}$ and of the definition of virtual displacements.

We can now apply Newton's second law $\overrightarrow{F}_i = m_i\ddot{\overrightarrow{r_i}}$ to each particle and take the virtual work of both sides. The sum over all particles leads to:

$$\sum_i m_i \ddot{\vec{r}}_i \cdot \delta\vec{r_i} = \sum_i \vec{F}_i^{\text{app}} \cdot \delta\vec{r_i} + \underbrace{\sum_i \vec{F}_i^{\text{c}} \cdot \delta\vec{r_i}}_{=0} \tag{9}$$

This yields to the famous d'Alembert principle:

$$\boxed{\sum_i \left(m_i \ddot{\vec{r}}_i - \vec{F}_i^{\text{app}} \right) \cdot \delta\vec{r_i} = 0} \qquad \textbf{D'ALEMBERT PRINCIPLE} \tag{10}$$

The constraint forces have disappeared entirely as they do no virtual work. What remains involves only the applied forces (which are known parameters) and the accelerations.

D'Alembert's principle converts Newton's law, which holds in the full ambient space and mixes known and unknown (constraints) forces, into a single scalar equation stated entirely within $\mathcal{C}$.

Remark It is crucial to remember that this holds for *holonomic* constraints (smooth surfaces, rigid rods, inextensible strings). Sliding friction violates this picture: it is a contact force with a tangential component, and therefore does nonzero virtual work.

Euler-Lagrange equation

As D'Alembert principle reformulate Newton's law within the configuration space, it is natural to express it not as a function of cartesian virtual displacements $\delta\vec{r_i}$ but generalized virtual displacements δq_α that acts as a basis of the configuration space. The two are related with the chain rule:

$$\delta\vec{r_i} = \sum_\alpha \frac{\partial \vec{r_i}}{\partial q_\alpha} \delta q_\alpha \tag{11}$$

D'Alembert principle thus rewrites as

$$\sum_{i,\alpha} \left(m_i \ddot{\vec{r_i}} \cdot \frac{\partial \vec{r_i}}{\partial q_\alpha} - Q_\alpha \right) \delta q_\alpha = 0 \tag{12}$$

With Q_α the forces expressed with generalized coordinates:

$$Q_\alpha = \sum_i \overrightarrow{F_i^{app}} \cdot \frac{\partial \vec{r_i}}{\partial q_\alpha} \tag{13}$$

As all the generalized coordinates q_α are a basis of the configuration space,the only way the above sum vanishes is if each term in the sum vanish separately (the δq_α form a free family of vectors in the configuration vector space). Therefore D'Alembert principle is rewritten as:

$$\sum_i m_i \ddot{\overrightarrow{r_i}} \cdot \frac{\partial \overrightarrow{r_i}}{\partial q_\alpha} = Q_\alpha \quad , \forall \alpha \tag{14}$$

As the cartesian coordinates depends both on generalized coordinates and time (cf Eq. 4), its total time derivative is obtained with the chain rule formula (we will explicitly writes $\dot{X} = \frac{d}{dt}(X)$ in the few next steps for clarity):

$$\frac{d\overrightarrow{r_i}}{dt} = \frac{\partial \overrightarrow{r_i}}{\partial t} + \sum_\alpha \frac{\partial \overrightarrow{r_i}}{\partial q_\alpha}\frac{\partial q_\alpha}{\partial t} = \frac{\partial \overrightarrow{r_i}}{\partial t} + \sum_\alpha \frac{\partial \overrightarrow{r_i}}{\partial q_\alpha}\frac{d q_\alpha}{dt} = \frac{\partial \overrightarrow{r_i}}{\partial t} + \sum_\alpha \frac{\partial \overrightarrow{r_i}}{\partial q_\alpha}\dot{q_\alpha} \tag{15}$$

The second equality comes from the fact that the $q_\alpha(t)$ have no explicit dependence on other q_β, so all partial derivatives collapse to total derivatives.. We'll note them $\dot{q}_\alpha$. From this last equation, we obtain by taking partial derivatives with respect to generalized coordinates and velocities the following:

$$\begin{cases} \frac{\partial}{\partial \dot{q_\alpha}}\left(\frac{d}{dt}(\overrightarrow{r_i})\right) & = \frac{\partial \overrightarrow{r_i}}{\partial q_\alpha} \\ \frac{\partial}{\partial q_\alpha}\left(\frac{d}{dt}(\overrightarrow{r_i})\right) & = \frac{\partial^2 \overrightarrow{r_i}}{\partial t \partial q_\alpha} + \sum_\beta \frac{\partial^2 \overrightarrow{r_i}}{\partial q_\alpha \partial q_\beta}\dot{q_\beta} = \frac{d}{dt}\left(\frac{\partial \overrightarrow{r_i}}{\partial q_\alpha}\right) \end{cases} \tag{16}$$

Let's use these two equations to derive Euler-Lagrange equation from D'Alembert principle ! First, let's rewrite the acceleration term using the product rule of derivatives:

$$\sum_i m_i \frac{d^2}{dt^2}(\overrightarrow{r_i}) \cdot \frac{\partial \overrightarrow{r_i}}{\partial q_\alpha} = \sum_i \left[\frac{d}{dt}\left(m_i \frac{d\overrightarrow{r_i}}{dt} \cdot \frac{\partial \overrightarrow{r_i}}{\partial q_\alpha}\right) - m_i \frac{d\overrightarrow{r_i}}{dt} \cdot \frac{d}{dt}\left(\frac{\partial \overrightarrow{r_i}}{\partial q_\alpha}\right)\right] \tag{17}$$

From Eq. 16 we can replace the partial derivative of $\overrightarrow{r_i}$ and $\dot{\overrightarrow{r_i}}$ with respect to q_α. The accelerration terms rewrites as:

$$\sum_i m_i \ddot{\overrightarrow{r_i}} \cdot \frac{\partial \overrightarrow{r_i}}{\partial q_\alpha} = \sum_i \left[\frac{d}{dt}\left(m_i \frac{d\overrightarrow{r_i}}{dt} \cdot \frac{\partial}{\partial \dot{q_\alpha}}\left(\frac{d}{dt}(\overrightarrow{r_i})\right)\right) - m_i \frac{d\overrightarrow{r_i}}{dt} \cdot \frac{\partial}{\partial q_\alpha}\left(\frac{d}{dt}(\overrightarrow{r_i})\right)\right] \tag{18}$$

Only one last step remain: notice that the velocities $\frac{d}{dt}(\overrightarrow{r_i})$ are functions of the generalized coordinates q_α and generalized velocities $\dot{q_\alpha}$ from eq. 4. As a consequence (using back the $\frac{dX}{dt} = \dot{X}$):

$$\begin{cases} \sum_i m_i \dot{\overrightarrow{r_i}} \cdot \frac{\partial}{\partial \dot{q_\alpha}} \dot{\overrightarrow{r_i}} = \frac{\partial}{\partial \dot{q_\alpha}}\left(\sum_i \frac{1}{2} m_i \dot{\overrightarrow{r_i}}^2\right) \\ \sum_i m_i \dot{\overrightarrow{r_i}} \cdot \frac{\partial}{\partial q_\alpha} \dot{\overrightarrow{r_i}} = \frac{\partial}{\partial q_\alpha}\left(\sum_i \frac{1}{2} m_i \dot{r_i}^2\right) \end{cases} \tag{19}$$

This makes the total kinetic energy $E_c = \sum_i \frac{1}{2} m_i \dot{r_i}^2$ appears! Finally, we can inject this new form of the acceleration into the rewritten D'Alembert principle:

$$\frac{d}{dt}\left(\frac{\partial}{\partial \dot{q_\alpha}}(E_c)\right) - \frac{\partial}{\partial q_\alpha}(E_c) = Q_\alpha \tag{20}$$

To finally get Euler-Lagrange equation, we should recall that some applied forces called "conservative" can be explicitly derived from a potential: $\overrightarrow{F_i^{app}} = -\partial_{\overrightarrow{r_i}} U$. For such forces, the generalized expression rewrites as:

$$Q_\alpha = \sum_i \overrightarrow{F_i^{app}} \cdot \frac{\partial \overrightarrow{r_i}}{\partial q_\alpha} = -\sum_i \frac{\partial U}{\partial \overrightarrow{r_i}} \cdot \frac{\partial \overrightarrow{r_i}}{\partial q_\alpha} + \sum_i \overrightarrow{F_i^{app,nc}} \cdot \frac{\partial \overrightarrow{r_i}}{\partial q_\alpha} = -\frac{\partial U}{\partial q_\alpha} + Q_\alpha^{nc} \tag{21}$$

Where $\overrightarrow{F_i^{app,nc}}$ and Q_α^{nc} denotes the applied forces and generalized forces that are not conservative and do not derive from a potential. Adding this decomposition in d'Alembert principle finally leads to Euler-Lagrange equation:

$$\boxed{\frac{d}{dt}\left(\frac{\partial}{\partial \dot{q_\alpha}}(\mathcal{L})\right) - \frac{\partial}{\partial q_\alpha}(\mathcal{L}) = Q_\alpha^{nc}} \qquad \textbf{EULER-LAGRANGE EQUATION} \tag{22}$$

The combination $E_c - U$ appears naturally because conservative forces contribute a potential term to both sides. We therefore introduced $\mathcal{L} = E_c - U$ the **Lagrangian** of the system.

Summary

At this stage, we can take a step back and appreciate what was achieved. Starting from a system of N particles in $\mathbb{R}^{3N}$, subject to k holonomic constraints and described by $3N$ coupled vector equations, we arrived at $d = 3N - k$ scalar equations expressed entirely in the generalized coordinates of $\mathcal{C}$. Three things happened simultaneously:

- The constraint forces (ie the unknown central difficulty of the Newtonian approach) disappeared entirely by construction, through the geometry of virtual displacements.
- The redundant coordinates were eliminated: the q_α are free and independent by definition, so no auxiliary conditions need to be carried alongside the equations of motion.
- The entire dynamics is encoded in a single scalar function $\mathcal{L} = E_c - U$: once the Lagrangian is written, the equations of motion follow mechanically by differentiation.

This is the power of the Lagrangian formalism: the physics is in the choice of $\mathcal{L}$; the rest is calculus.

Remark: Scleronomic and rheonomic constraints

The derivation above assumed implicitly in Eq. 19 that the constraint equations do not depend explicitly on time, i.e. $f(\overrightarrow{r}_1, \ldots, \overrightarrow{r}_N) = 0$ with no t dependence. Such constraints are called **scleronomic** (from the Greek *skleros*, rigid). When constraints do depend explicitly on time $f(\overrightarrow{r}_1, \ldots, \overrightarrow{r}_N, t) = 0$, they are called **rheonomic** (from the Greek *rheos*, flow).

The Euler-Lagrange equation remains valid in both cases. In our derivation, eq. 15 was correct both for rheonomic and scleronomic case (where the $\frac{\partial \overrightarrow{r}_i}{\partial t}$ term vanishes). However, we assumed we were working with scleronomic constraints when we introduced the kinetic energy in Eq. 19. Indeed, expanding $E_c = \sum_i \frac{1}{2} m_i \dot{\overrightarrow{r}}_i^2$ then produces terms as $\partial_t \overrightarrow{r_i} \cdot \partial_{q_\alpha} \overrightarrow{r_i} \dot{q}_\alpha$ (linear in $\dot{q_\alpha}$) and $|\partial_t \overrightarrow{r_i}|^2$ (constant in $\dot{q_\alpha}$), in addition to the usual quadratic terms. As a consequence, the lagrangian $\mathcal{L} = E_c - U$ is no longer a purely quadratic form in $\dot{q}_\alpha$, and several results that rely on this homogeneity (for instance the identification of E_c with the total mechanical energy) no longer hold without modification. In what follows, we'll detail the consequences of rheonomic constraints on Euler-Lagrange equation as well as their properties.

Remarkably, this same equation will re-emerge from a completely different starting point: the principle of least action, which recasts the entire problem as the search for a path that extremizes a scalar functional. This equivalence is not a coincidence: it reflects a deep geometric structure that will become central in the transition to quantum mechanics.

1.1.3 Properties of Euler-Lagrange equation, action and Variational principle

soon

1.1.4 Hamilton equations

References